

Subtract 00:07:20 from timestamps to find place in audio/video recording.

18

00:07:23.000 --> 00:07:24.109

Zargham: Can you hear me? Okay.

19

00:07:25.790 --> 00:07:27.009

Jamsheed Shorish: Yeah. Can hear it.

20

00:07:28.360 --> 00:07:42.636

Zargham: I'm at. I'm in Bloomington, but I needed to do some work, so I stayed back at the Airbnb for the morning, so I figured I'd I join you guys, I'm preparing my slides tomorrow I give a talk about attention economies and governance services.

21

00:07:43.490 --> 00:07:44.370

Jamsheed Shorish: Oh, cool!

22

00:07:45.730 --> 00:08:05.590

Zargham: I also sent of an email Ccd or connected you in but a presentation yesterday from Seth Fry and one of his collaborators, I think covered some really interesting modeling and I invited them to kinda coordinate with us to give a guest lecture around

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00:08:06.150 --> 00:08:16.719

Zargham: like how you model agents, and the kind of sort of institutional rules, and some some other, I think, really relevant stuff that people will enjoy. So I fired off.

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00:08:17.500 --> 00:08:20.390

Zargham: Seth, have made the connection, so you'll see that.

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00:08:21.110 --> 00:08:24.967

Jamsheed Shorish: Great. Then I'll coordinate with them about setting up one of our open slots to have and give a talk.

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00:08:25.380 --> 00:08:30.589

Zargham: Perfect. Yeah, it's actually, I think if you guys will like like it, anyway.

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00:08:31.730 --> 00:08:32.340

Jamsheed Shorish: Thanks.

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00:08:35.840 --> 00:08:37.339

Jamsheed Shorish: I think we have a pretty good

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00:08:37.850 --> 00:08:44.040

Jamsheed Shorish: schedule up through. Yeah, it'd be toward the end of the summer, and it should be. Then we have an open slate for the fall, so.

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00:08:44.049 --> 00:08:56.049

Zargham: Yeah, I figured it would be no a no rush kind of thing but it it was nice to you know. I I would like to see us have a little a more cross links with people who are like kind of not

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00:08:56.129 --> 00:09:04.499

Zargham: primarily interested in crypto problems, but for which their research is relevant to sort of modeling generally. So.

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00:09:04.500 --> 00:09:05.390

Jamsheed Shorish: Definitely.

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00:09:05.980 --> 00:09:08.255

Jamsheed Shorish: I was just having a conversation with

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00:09:09.160 --> 00:09:12.130

Jamsheed Shorish: with octopus, in fact, about the same thing, about

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00:09:12.680 --> 00:09:16.490

Jamsheed Shorish: under what conditions and how we should be thinking about leveraging

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00:09:17.040 --> 00:09:18.909

Jamsheed Shorish: for lack of a better word, sort of

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00:09:19.190 --> 00:09:25.969

Jamsheed Shorish: taxonomies of agents, kind of like what we had done with a graph before, so I gave them some of that earlier work that we had been talking about a little bit.

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00:09:26.190 --> 00:09:27.599

Jamsheed Shorish: and and

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00:09:27.750 --> 00:09:35.909

Jamsheed Shorish: see whether or not that makes sense, because he's thinking about that within the terms, in terms of the the maids, applications within Cadcad and via Gds

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00:09:36.360 --> 00:09:45.749

Jamsheed Shorish: to see how that might be. A nice way to parameterize sort of parameterized agents in essence to extend the parameter space that you're looking at to encompass types.

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00:09:45.750 --> 00:10:09.383

Zargham: Yeah. And I I think the thing I I don't wanna spend too much time cause I know we have a topic for today. But I the thing I liked about that work was that they had a kind of structural model where the agents were learning their policies. And so, like, you know, seeing environments where obviously you have to have a some of a structural model in place for them to learn policies. But it was more open ended like what the agents could learn to do.

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00:10:09.927 --> 00:10:19.424

Zargham: Wasn't just learning decisions, but actually learning policies. And for that that was kind of the tipping point where I was like, Aha! We should talk to you more because it was

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00:10:20.000 --> 00:10:20.920

Carolina: Mario.

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00:10:21.631 --> 00:10:23.039

Carolina: This is an RL. Model.

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00:10:23.100 --> 00:10:24.739

Carolina: but Enforcement learning model.

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00:10:25.206 --> 00:10:47.389

Zargham: It was not quite. I would say it wasn't quite a full on RL model. It's interesting because I have mixed feelings about the RL. Literature, because I think a lot of the times. It squashes some of

the internal structural models that matter. But obviously, if you impose a structural model and then do reinforcement learning. You're learning the coefficients of the structural model that you have.

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00:10:47.390 --> 00:11:12.339

Zargham: So it was a relatively short presentation. So we're mostly seeing results and insights and kind of future questions. But it was institutional and like agent modeling focus. So in particular, for us, where I think we have interest in analyzing not purely financial decisions and kind of markets and market structures over objects that are not just prices. And that was something we had, in common with those reasons.

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00:11:12.340 --> 00:11:13.050

Zargham: searchers.

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00:11:13.790 --> 00:11:20.880

Carolina: Okay, that's great. And what was the underlying model? Then what was underline material use to learn these policies.

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00:11:21.183 --> 00:11:24.213

Zargham: It was a queue learning model, I think, which is

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00:11:25.535 --> 00:11:32.644

Zargham: But yeah, I'd have to go back and check the the source material, but ideally we'll have the authors come, speak with us, and we can ask them.

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00:11:33.540 --> 00:11:35.017

Carolina: Perfect. That's great.

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00:11:36.170 --> 00:11:40.108

Carolina: Some people would argue, though, that Q. Learning is is unfortunate learning, but anyways.

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00:11:40.390 --> 00:11:45.440

Zargham: Yeah, I don't. Yeah. And I think that's a that. I agree. I think the challenge is the

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00:11:45.700 --> 00:11:56.622

Zargham: reinforcement. Learning has gotten kind of squished out where

people use it to refer to lots of stuff, and that's fine. But like, I think, on a narrower taxonomy, I think I would say it's Q learning.

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00:11:57.740 --> 00:11:58.879

Carolina: Sounds good. Yeah, yeah.

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00:11:59.480 --> 00:12:04.099

Jamsheed Shorish: I think we are good to go with quorum, so I'll jump in quickly here.

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00:12:04.838 --> 00:12:06.781

Jamsheed Shorish: But everybody can see that

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00:12:07.390 --> 00:12:08.480

Jamsheed Shorish: title slide.

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00:12:08.800 --> 00:12:09.480

Carolina: Yes.

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00:12:10.160 --> 00:12:30.529

Jamsheed Shorish: Great. So this is actually a an application of work that's currently in progress. That we're doing. Actually, Carolina and I are putting together something that's an ask from Aztec. That essentially, is the way in which they would like to adopt a particular kind of voting or consensus mechanism for submitting proofs of transactions.

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00:12:30.550 --> 00:12:52.500

Jamsheed Shorish: just as a preclin just sort of disclaimer right at the beginning. It's all nda. So you can't, you know, go around spreading this type of stuff. But of course, it's just more about the discussion from what's useful from our perspective, about how do we approach the problem, what is the problem statement? What is it that we think is a a cohesive method of building a model, especially within a very short timeframe? Because this is actually just 2 weeks worth of worth of time available.

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00:12:52.910 --> 00:13:12.880

Jamsheed Shorish: So just with that preamble, what we're looking at here is under Aztecs environment, just to give a quick reiteration of what Aztec is. And of course Jacob can jump in and slam me if I'm not getting giving them the the do service that they deserve. But it's an

L. 2. Networking service for public and private transactions that use  
l. One for their global security.

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00:13:13.210 --> 00:13:24.079

Jamsheed Shorish: And what they have is a particular randomization procedure for proposing an individual. One of the sequencers, as they're called to propose a series of transactions to be included in a block

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00:13:24.290 --> 00:13:38.739

Jamsheed Shorish: that then is offloaded beyond, either behind the scenes with the sequencer, or through a 3rd party, prover, to actually submit a proof that is essentially rolled up, then on the layer, 2 to then be submitted as a transaction itself on layer one.

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00:13:39.650 --> 00:14:07.609

Jamsheed Shorish: the current security model that they have for the proving process, not for the entire security of the system, but really about whether or not the data that's coming in as the transactions in the block is valid, there. There actually are valid transactions. Currently their model, they're thinking around the model is to have the prover essentially acting as putting up a stake, a commitment bond, as it's called, and essentially that is available to be slashed in the event that a proof is not forthcoming.

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00:14:07.990 --> 00:14:27.749

Jamsheed Shorish: and that that slashing that occurs once that happens. Everybody says, Okay, well, now, it's a free for all. We don't have a proof for the existing proposed block. Therefore there's going to be a race mode that's being triggered where it's the 1st pass the post. Anybody who has a particular valid block can now receive the and submit a proof, get it finalized, and receive any kind of reward that's associated with that.

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00:14:28.930 --> 00:14:35.470

Jamsheed Shorish: And the intuition here, which is our web. 3. Intuition is an idea that, hey? If you stake a lot, then there's a lot more at risk.

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00:14:35.530 --> 00:14:56.770

Jamsheed Shorish: And therefore there's greater economic security in the sense that it would be more expensive to compensate that risk by taking over the individuals who are doing these things in order to, for example, trigger race mode deliberately and try to earn some

returns so very quick nutshell of what it is that the Aztec system, as it currently stands, is, is set up for.

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00:14:57.390 --> 00:14:59.029

Jamsheed Shorish: The thing that they're worried about

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00:14:59.040 --> 00:15:22.190

Jamsheed Shorish: is that this commitment bond itself is potentially a weak incentive for doing what they would like it to be able to do. 1st of all, it may be just simply too high a bond amount for many of the sequences improvers which ends up centralizing the system for large entities initially for bootstrapping. That may be fine. Ultimately they would like to have this as decentralized as possible. And so this may be a barrier to decentralization.

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00:15:22.930 --> 00:15:39.480

Jamsheed Shorish: Second, the time that it takes to finalize a proof may itself be quite long, because you may actually have these very large, complicated proposal for blocks, and then the requirements of the system to be snappy and kind of allow many blocks to be processed goes down. So you have less liveness

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00:15:39.560 --> 00:15:40.729

Jamsheed Shorish: in the system.

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00:15:41.600 --> 00:15:50.219

Jamsheed Shorish: Now, of course, in the end there may be the case that the entity itself is incentivized to not prove a block by triggering race mode or something else.

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00:15:50.270 --> 00:16:01.109

Jamsheed Shorish: that profitability itself is connected very much to the l 1 entities that actually finalize the transactions and the roll ups, because that's the way in which you actually can get paid once. That's been finalized on l 1.

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00:16:01.551 --> 00:16:20.210

Jamsheed Shorish: That's going to be the responsibility of other mechanism modifications. So we won't treat with that particular discussion here, but it's worth mentioning that there is always that other side. There's a cost involved, and there's also a potential benefit involved. And what's being asked for here is to fulfill the role of making it expensive

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00:16:20.330 --> 00:16:24.729

Jamsheed Shorish: to deviate from the intended behavior in this commitment bond environment.

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00:16:26.010 --> 00:16:43.889

Jamsheed Shorish: So what we really are looking for then, and what they're looking for is investigating whether or not there's another approach that can increase this cost of not proving a block proposal, but at the same time on the margin, hope that it doesn't actually impact liveness, or, in fact, reinforces the ability to be snappier or more responsive in a sense.

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00:16:44.040 --> 00:16:49.100

Jamsheed Shorish: And so they are currently debating. And they're asking for us to provide input for

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00:16:49.130 --> 00:16:54.450

Jamsheed Shorish: a consensus mechanism for the block data for the proving process

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00:16:54.480 --> 00:17:06.039

Jamsheed Shorish: and what they want to do here is to help incentivize, then, the ability of when the time period is coming to the end of the period in which a proof should be available. A proof is there to be processed and used.

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00:17:06.940 --> 00:17:31.139

Jamsheed Shorish: So the idea here is, and they're kind of leveraging a little bit of what tendermint, I think, has been doing is to look for a pool of validators that are nominated from the entire set of the sequencers. So the sequences are randomly generated. Every so often there's a lead sequencer that proposes a block, but the remaining pool still gets to interact, and so randomly they can actually be collected or selected as a pool of validators to a test

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00:17:31.280 --> 00:17:41.330

Jamsheed Shorish: that the data of the proposed block from the lead sequencer is valid, in other words, that eventually, when push comes to shove that, it is provable that it can be done.

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00:17:42.390 --> 00:17:52.580

Jamsheed Shorish: And the idea here is that you put up again a kind of a commitment bond. Here. It's just called a stake. The sequencer's stake for participation becomes the thing that is at risk

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00:17:52.680 --> 00:18:00.160

Jamsheed Shorish: from the validator that that's their commitment to being able to submit that or attest to this ability of the data.

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00:18:00.910 --> 00:18:03.070

Jamsheed Shorish: And if an attestation occurs.

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00:18:03.190 --> 00:18:10.360

Jamsheed Shorish: there's some period of time that's presupposed and prespecified, within which the time which that proof itself can be submitted.

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00:18:10.420 --> 00:18:11.470

Jamsheed Shorish: And

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00:18:11.500 --> 00:18:21.930

Jamsheed Shorish: and this is the idea to improve liveness, during which time people can say, Oh, that's great. You got that proposal. We're pretty sure you guys have attested to that proposal. Fantastic. I'm going to build on that block.

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00:18:22.090 --> 00:18:25.230

Jamsheed Shorish: And so you then have the succeeding blocks being built

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00:18:25.360 --> 00:18:27.660

Jamsheed Shorish: before the proof has been submitted.

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00:18:27.800 --> 00:18:39.939

Jamsheed Shorish: So the attestation is acting as sort of a green flag for the ability to have optimistic look ahead, and that's going to in in the theory and the principle of the idea, improve the the responsiveness or the liveness of the ecosystem.

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00:18:41.590 --> 00:19:00.840

Jamsheed Shorish: Now, if a proof in the end isn't submitted. Then, of course, there is the expense that's associated with this. The

validators themselves have to prove that it was possible to do so, so that they're not actually being left on the hook, and they weren't actually leading everybody astray. So they have to be able to say, hey, look! There's another reason why that proof didn't show up. It's not us. We actually attest it to correct data.

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00:19:01.010 --> 00:19:08.200

Jamsheed Shorish: But in the event that that's not the case if they can't support and show that that data is there, or that that data is itself invalid.

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00:19:08.410 --> 00:19:14.990

Jamsheed Shorish: then the validators were slashed. Otherwise they're punished for having provided an attestation for something that didn't materialize.

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00:19:15.020 --> 00:19:25.140

Jamsheed Shorish: The stake of the lead sequencer itself is also slashed, because, ostensibly there may have been missing metadata or other data that was necessary for that block to have been proven, that data to have been proven.

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00:19:25.190 --> 00:19:31.369

Jamsheed Shorish: and finally, the chain is reorganized back to the original proposal time slot. So you have to unwind. Everything had happened before.

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00:19:31.470 --> 00:19:40.949

Jamsheed Shorish: So it's an expensive idea, expensive for the protocol, of course, in the event that it happens, but expensive to the validators if they turned out to have not attested properly.

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00:19:42.290 --> 00:19:43.150

Jamsheed Shorish: Okay.

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00:19:43.500 --> 00:19:50.439

Jamsheed Shorish: before moving on, I just wanna make sure that that's sort of clear for everybody any questions about this sort of compare and contrast between these 2 approaches.

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00:19:53.490 --> 00:19:54.080

Carolina: Soup.

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00:19:54.600 --> 00:19:55.240

Jamsheed Shorish: Okay.

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00:19:55.650 --> 00:20:03.180

Hash: These validators that you spoke of? Are these the validators on the sort of this relay chain? Or is it the set of validators that are also doing

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00:20:03.360 --> 00:20:04.650

Hash: like sequencing

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00:20:05.210 --> 00:20:06.540

Hash: like ugly. It's not.

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00:20:06.540 --> 00:20:13.110

Jamsheed Shorish: They are sequencers but themselves. They are selected as validators, if they have not been selected as a sequencer to propose a block.

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00:20:13.780 --> 00:20:14.957

Hash: Oh, okay. Thanks.

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00:20:15.350 --> 00:20:24.850

Zargham: So these are 2 different roles, and the discussion is the extent to which the same entity can potentially hold them on, and the conditions under which they can hold them.

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00:20:25.910 --> 00:20:29.509

Jamsheed Shorish: Yeah, exactly. And they aren't on l 1. They're on the layer. 2 chain.

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00:20:31.130 --> 00:20:36.180

Hash: The when when they are settling the transactions on the layer sort of this relation, or the layer one.

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00:20:36.930 --> 00:20:46.149

Jamsheed Shorish: So they're sending whether or not the data itself is valid, that's all that's all they're doing. They're discussing whether or not the data itself can be proven, and the idea is to incentivize

the submission of a proof.

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00:20:46.340 --> 00:20:56.079

Jamsheed Shorish: Then there is the point where the actual proof is rolled up and sent over to L. One, where the security of L. One is used to finalize the series of transactions.

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00:20:57.580 --> 00:20:58.300

Hash: Thanks.

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00:20:58.300 --> 00:20:58.960

Jamsheed Shorish: Okay?

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00:20:59.340 --> 00:21:15.099

Jamsheed Shorish: So the ask from Aztec was, Hey, in these 2 weeks? So starting from last almost a week, a week ago, tomorrow, could you develop a way to kind of like, define for us a number, a magic number for economic security of this consensus mechanism, as defined with

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00:21:15.320 --> 00:21:28.639

Jamsheed Shorish: lots of caveats on the basis of of what's going on, but what they would like to have these requirements of this ask are to have it comparable to other mechanisms, so that they can actually compare the commitment bond mechanism, for example, with the consensus mechanism that they have.

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00:21:28.990 --> 00:21:55.330

Jamsheed Shorish: it has to be fairly high level mainly due to time constraints. So we don't want any behavioral assumptions specifically that make it too difficult to generalize. They don't want to go into the weeds that we don't want to go into the weeds of the technical implementation of the consensus mechanism itself. It should be something that can be treated, let's say at an abstract level, but still provide in the end meaningful dependencies between the parameters that they need to select for the consensus mechanism

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00:21:55.490 --> 00:21:59.129

Jamsheed Shorish: and what the outcomes of with regard to this particular measurement.

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00:21:59.530 --> 00:22:05.729

Jamsheed Shorish: So we're not doing any kind of mechanism design.

What we're doing here is really trying to surface a metric

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00:22:05.820 --> 00:22:08.030

Jamsheed Shorish: that is consistent with

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00:22:08.100 --> 00:22:10.679

Jamsheed Shorish: these requirements that they're asking us to perform.

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00:22:12.040 --> 00:22:29.970

Jamsheed Shorish: And so our solution, which is currently underway, the 1st part of which I'll show you, is the lightweight model that fulfills these above requirements. And it's lightweight. So you're gonna see, you're gonna see lots of assumptions. And you're gonna say, that's a crazy assumption. But it's a lightweight model that still provides insight into some measurement of what we might call economic security.

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00:22:30.481 --> 00:22:56.359

Jamsheed Shorish: And in addition to this, there's also a tool which is currently Carolina is currently already developing as a means of basically changing this sort of like, what if scenarios, if I change particular consensus parameters. I observe the resulting change in the economic security. This allows me to change and compare it not just the commitment bond to the consensus mechanism under one set of priorities or one set of parameters, but also to compare them between different sets of assumptions.

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00:22:59.370 --> 00:23:00.150

Jamsheed Shorish: Barrett.

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00:23:00.650 --> 00:23:23.070

Jamsheed Shorish: 1st thing you need is a definition. So we want to try and find something that we can say. This is our very general case of how we specify a metric of measurement. And so a very lightweight definition of economic security which is in essence, you think about this sort of as the hypothesis that you later design the system to test, although, of course, you're doing both at the same time, you're thinking, okay, how would I calculate this at the same time as you're specifying it?

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00:23:23.260 --> 00:23:26.449

Jamsheed Shorish: But an abstract definition of economic security

would be.

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00:23:26.460 --> 00:23:28.120

Jamsheed Shorish: well, what's at stake?

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00:23:28.350 --> 00:23:36.249

Jamsheed Shorish: What is it that's at risk from the individuals in much the same way that the commitment bond placed something at risk. What you're looking for is the expected loss

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00:23:36.760 --> 00:23:39.609

Jamsheed Shorish: to a validator from being involved

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00:23:39.760 --> 00:23:48.429

Jamsheed Shorish: in essentially a a bad situation. The joint event that an attestation was provided there was like, say, Yes, go for it. This data is good to go.

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00:23:48.510 --> 00:23:52.710

Jamsheed Shorish: but a proof was unavailable at the time that it was supposed to be due. Nothing showed up.

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00:23:52.870 --> 00:24:00.389

Jamsheed Shorish: and then, when they were called to account, the validator could not produce the data, or it turned out oops. The data itself was invalid.

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00:24:02.600 --> 00:24:09.440

Jamsheed Shorish: That's a very general definition. It allows us to say what's at risk by relying upon the probability of loss

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00:24:09.540 --> 00:24:10.830

Jamsheed Shorish: as a validator.

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00:24:11.210 --> 00:24:12.160

Jamsheed Shorish: and

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00:24:12.430 --> 00:24:27.849

Jamsheed Shorish: we have a lot of degrees of freedom. So the benefit of this also is that you can pair down lots of things for this 2 week

horizon that you can then say, you know, work with us to generate a create, a general model of this, and we can actually build much more realistic assumptions around this, because the definition doesn't change

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00:24:27.890 --> 00:24:48.749

Jamsheed Shorish: only the things that we're assuming underneath change. If you make your definition of your metric too restrictive, then, if somebody says, Yeah, but you assume this, and you go. Oh, I have to break that definition. Now then, you've lost something, whereas if all you need to do is under the hood, change the assumptions, derive the right statistics, and then say it's the same kind of measurement. Then you've actually kept yourself kind of ahead of the game in a way.

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00:24:49.100 --> 00:24:58.739

Jamsheed Shorish: But in this environment we're going to use independence of certain types of events as our Ockham's razor to slice things through to get things underway over this two-week horizon.

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00:24:59.910 --> 00:25:05.030

Jamsheed Shorish: This gives us enough complexity to still do that nice parametric dependency example.

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00:25:05.160 --> 00:25:16.540

Jamsheed Shorish: but it's simple enough that we can get some nice closed form modeling. What I mean by close form modeling is in the sense that it can be implemented in a spreadsheet in the tool that is deliverable, so we don't have to say

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00:25:16.560 --> 00:25:30.710

Jamsheed Shorish: can't do it in a spreadsheet. We have to put this into, you know, a catcap framework. We have to put it into a python simulation or some other method that is less user friendly and is less what they're asking for in terms of this, this tool that just lets them tinker with parameters in a visual format.

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00:25:33.330 --> 00:25:54.799

Jamsheed Shorish: So the very simple independence assumptions. And I'll start to show you some more of the definitions before wrapping up toward the end. If we have time, of course, and interject questions at any time, but if we have time, I'll show you some extensions that are also going to go into the final tool at the end. But the very beginning idea is just to say, Okay, let's think about this space of

random things that could occur. Call that for the usual standard capital, Omega.

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00:25:55.080 --> 00:25:56.750

Jamsheed Shorish: And let's think about the

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00:25:56.810 --> 00:26:04.040

Jamsheed Shorish: following 3 different types of events. The 1st one is that the event the data is attested to by the Validator Pool.

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00:26:04.340 --> 00:26:13.709

Jamsheed Shorish: Now you might think, well, that's deterministic. People see the thing that they're getting, and then they make a decision they vote on whether or not to actually say this data is valid.

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00:26:13.720 --> 00:26:18.950

Jamsheed Shorish: But from the point of view of the arrival of future blocks, the blocks that are coming online.

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00:26:19.110 --> 00:26:22.420

Jamsheed Shorish: You don't know what the outcome of the voting will necessarily be.

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00:26:22.520 --> 00:26:29.520

Jamsheed Shorish: So you consider this to be a random, variable, and say, well, the data is attested by the Validator Pool with some probability.

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00:26:30.110 --> 00:26:35.859

Jamsheed Shorish: There's another event which is the same thing about the really reflecting the idea of the deliverability of the proof

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00:26:35.940 --> 00:26:37.850

Jamsheed Shorish: that when the time comes

151

00:26:37.860 --> 00:26:40.720

Jamsheed Shorish: the proof is either available or it is not.

152

00:26:41.530 --> 00:26:42.710

Jamsheed Shorish: And finally.

153

00:26:42.730 --> 00:27:03.370

Jamsheed Shorish: once you see that the hey? This data itself hasn't shown up? Well, then, you have to actually go back and say, Is it possible for the validator to show and prove. Hey, look! I attested to something that was valid. And so there is an event under which you would say, Oh, can't do it. The data itself was incomplete or unprovided. In that case, from the Validator Pool.

154

00:27:04.140 --> 00:27:17.020

Jamsheed Shorish: and the joint probability of sort of being in the worst mess is the joint probability of these 3 events occurring which can be pretty complex, you can think of a lot of different ways in which the structure of the block proposal

155

00:27:17.130 --> 00:27:33.079

Jamsheed Shorish: impacts the way in which the attestation occurs which impacts or itself is impacting the way in which the proof can be done which may also have an impact on whether or not you could actually replicate. It turns out that you actually validated something that was invalid, for example.

156

00:27:34.380 --> 00:27:48.389

Jamsheed Shorish: Well, let's start pairing down in our usual way. Let's break our joint distribution into nice conditional distributions so that we can productize these distributions. We'll say, well, there's a probability that the data is attested to conditional

157

00:27:48.450 --> 00:27:56.639

Jamsheed Shorish: upon it being unavailable or being incomplete. And then that's multiplied by that joint distribution of that bad event that happens at the end.

158

00:27:57.030 --> 00:27:59.500

Jamsheed Shorish: And here's our cut, our 1st cut.

159

00:27:59.600 --> 00:28:01.470

Jamsheed Shorish: We say, all right, let's just

160

00:28:01.730 --> 00:28:09.270

Jamsheed Shorish: dump the dependencies on the proof being unavailable

or the data being incomplete. And just say that there is a probability of attestation.

161

00:28:09.930 --> 00:28:17.490

Jamsheed Shorish: and it's independent of anything else, and we'll be specific about what that means and how rich that space can still be with regard to individuals.

162

00:28:17.810 --> 00:28:26.230

Jamsheed Shorish: but this significantly reduces the complexity at the expense of being realistic. So we can't necessarily get good correlations based on that

163

00:28:26.840 --> 00:28:42.440

Jamsheed Shorish: in a full model. And of course this is one of the things I want to harken back to some of the the if you guys have been following the seminar series, that octopus has been putting out this links tightly back to the idea of well, hey, look! We could actually build behaviors about individuals and what they're conditioning on what their information is

164

00:28:42.490 --> 00:28:53.840

Jamsheed Shorish: that would enable us to have a very rich structure for that probability, that joint distribution. So you can think about actually in one direction of extension would be in that maids style direction.

165

00:28:56.000 --> 00:28:56.820

Jamsheed Shorish: Okay?

166

00:28:57.030 --> 00:29:03.990

Jamsheed Shorish: So we've got this voting mechanism. Now, that is going to be built upon this probability that an attestation is made

167

00:29:04.310 --> 00:29:09.440

Jamsheed Shorish: of the data of this proposed block. Given that it's independent of the other events.

168

00:29:10.120 --> 00:29:13.709

Jamsheed Shorish: So let's start out with a pool of S sequencers

169

00:29:13.880 --> 00:29:38.290

Jamsheed Shorish: something in the round. I think they've been talking about things as high as 5,000 thousands of potential sequences in a sequence or pool could be hundreds could be thousands. There's N validators. There apparently is a hard limit of 300 as a maximum number of validators, but it could be a certain number of validators that are selected each time a block is proposed. There's a pool of invalidators. These numbers don't change over the course of the of the assessment.

170

00:29:39.060 --> 00:29:43.600

Jamsheed Shorish: and as an assumption we would make the assumption. Well, let's think about this probability, then.

171

00:29:43.730 --> 00:29:45.429

Jamsheed Shorish: based upon

172

00:29:45.480 --> 00:29:53.150

Jamsheed Shorish: if you're a validator looking at the rest of your pool of validators, and implicitly assuming that you're not a clone, and you're not actually all the same individual.

173

00:29:53.340 --> 00:30:00.699

Jamsheed Shorish: Now, there's a probability that any particular validator votes to a test, a blocks, data being the number, a

174

00:30:01.440 --> 00:30:08.959

Jamsheed Shorish: some probability, some fraction. The the voting, from your point of view, looking at the rest of the universe, the rest of your your validators. It's random

175

00:30:09.360 --> 00:30:20.619

Jamsheed Shorish: now, as a relaxation, we can relax, and we have. So there's results that I can show at the very end where each validator actually has their own probability. So everybody sort of idiosyncratic.

176

00:30:20.880 --> 00:30:32.580

Jamsheed Shorish: This is more realistic. It has a little slightly higher complexity associated with it, and it doesn't necessarily throw out the entire problem of how do you understand the distribution of your fellow peers as validators.

177

00:30:32.590 --> 00:30:45.060

Jamsheed Shorish: But at 1st blush, we're gonna make the sort of the heroic assumption that everyone has. The same probability. It doesn't mean the same outcomes will occur. Just means we all have the same chance of voting to attest the data of a block.

178

00:30:46.130 --> 00:30:46.990

Jamsheed Shorish: Okay.

179

00:30:47.490 --> 00:30:56.450

Jamsheed Shorish: so what's this probability that an attestation is made once the data has come in. Well, it's the probability that the total number of votes to attest the data

180

00:30:56.610 --> 00:30:58.299

Jamsheed Shorish: is a majority.

181

00:30:58.420 --> 00:31:19.680

Jamsheed Shorish: So here we're just using the ceiling function here to accommodate an even or an odd number of N. And as Jacob De pointed out quite early, hey, look, you know we can. We can just use the 50% plus one tie breaking rule. And that will give us the evens in the stuff that follows here. I'm just going to be assuming without loss of generality that N is odd just because it makes it easier. I can drop the ceiling functions and

182

00:31:19.730 --> 00:31:34.640

Jamsheed Shorish: just work with numbers that are that are easily specified. So, for example, that  $n$ , minus one over 2 plus one, then becomes majority vote in an odd number of of see of validators. And that's just  $n$  plus one over 2.

183

00:31:37.090 --> 00:31:37.830

Jamsheed Shorish: Okay?

184

00:31:38.480 --> 00:31:54.190

Jamsheed Shorish: Looks okay. So far, we've got the idea of what the threshold should be that  $n$  plus one over 2 is the threshold the votes have to exceed. We now know that we can write something like this. We should be able to figure out a way to combine that with this probability,  $a$ . That any validator votes to a test a block.

185

00:31:54.890 --> 00:32:01.819

Jamsheed Shorish: And, sure enough, what you end up with is the number of you can think of each of these votes as a success or a failure.

186

00:32:01.850 --> 00:32:09.860

Jamsheed Shorish: Success means it was voted in favor of. The rejection is a failure. And so what you're actually following is a binomial distribution in votes.

187

00:32:10.450 --> 00:32:28.380

Jamsheed Shorish: So you could say, Well, great then, the the tail of the distribution, that part which is just majority voting majority voting is nothing more than the number of possible ways in which I could combine the individuals to get probabilistically. The mass to go over that particular threshold of  $N$ . Plus one over 2,

188

00:32:28.870 --> 00:32:46.069

Jamsheed Shorish: and that's that right hand summation where the the 2 numbers in the in parentheses the  $N$  choose.  $K$  is just  $n$  factorial over  $K$  factorial times,  $n$ , minus  $K$  factorial. So it's a number of possible combinations that I could have selected  $K$ . From the population of sequence of validators.  $N$ .

189

00:32:46.240 --> 00:32:51.779

Jamsheed Shorish: And that gives you the number of combinations to make sure you're properly counting the possible ways. You could have gotten a majority vote.

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00:32:52.900 --> 00:32:53.640

Jamsheed Shorish: So

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00:32:53.990 --> 00:32:56.190

Jamsheed Shorish: at this stage, you're kind of like, okay, that's

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00:32:56.370 --> 00:33:22.569

Jamsheed Shorish: nice, maybe. But that's actually hard to compute. Numerically, it's quite expensive to do in and choose  $K$  to actually do this in an exact form. There's a lot of literature in the computer science literature, especially on approximating this, because it comes up again and again and again in stochastic possibilities. We've we're using normal approximations. We can use churn of bounds. I'm gonna abstract away from those details, although they are interesting and

simply say there exists a bound or an approximation

193

00:33:22.570 --> 00:33:35.080

Jamsheed Shorish: that we can use. And we're gonna approximate. Then this probability of the attestation occurring by some approximation.  $P$ . Sub  $b$ , and that's gonna depend upon the number of sequencers sorry the number of validators. And

194

00:33:35.140 --> 00:33:36.650

Jamsheed Shorish: and this probability

195

00:33:36.660 --> 00:33:37.969

Jamsheed Shorish: that you accept a vote.

196

00:33:40.110 --> 00:33:48.259

Jamsheed Shorish: Now, why do we care about this? Because we're looking for this value of the expected loss. So when a loss has occurred.

197

00:33:48.330 --> 00:34:03.629

Jamsheed Shorish: It's when the data has been attested. But the proof is not available and the data was not provided. X post. So we need this component. That's that  $P$ . Of  $\Omega$   $a$ . We're just rewriting it now as an approximation of the actual voting process that's being set up for this consensus mechanism.

198

00:34:04.640 --> 00:34:09.880

Jamsheed Shorish: That means, of course, that the more likely the data is being attested to all other things being equal.

199

00:34:10.080 --> 00:34:11.559

Jamsheed Shorish: the higher the loss.

200

00:34:12.550 --> 00:34:26.739

Jamsheed Shorish: It may seem counterintuitive to think about the idea of Hey, I'm doing my part as a good citizen to be in the voting process. But the problem is that if there are negative consequences associated with certain outcomes of your voting. Then that could be something that when that actually happens, you feel worse.

201

00:34:27.260 --> 00:34:33.599

Jamsheed Shorish: So that's part of where we get the economic security from again. This is this notion that you've got something at risk.

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00:34:36.699 --> 00:34:56.040

Jamsheed Shorish: Okay? And just to reiterate and underscore, we're not assuming anything about the validators behavior there would be ample room and opportunity to do so. It would be a nice way to say that there is, as Sargan was just mentioning, like a taxonomy of agents that we could put into the system and have. We would have a 2, you know, turnkey solution to look at different types

203

00:34:56.320 --> 00:35:07.840

Jamsheed Shorish: of outcomes on the basis of voting behaviors. Here. Very short horizon we just keep with the probabilistic approach of whether or not an individual validator is approving or attesting to this data.

204

00:35:10.630 --> 00:35:11.370

Jamsheed Shorish: Okay.

205

00:35:12.050 --> 00:35:14.770

Jamsheed Shorish: with this in hand, we can now create

206

00:35:14.780 --> 00:35:37.369

Jamsheed Shorish: a measure, a statistic. And this is where I'm calling it the economic security stake. And we're thinking about this at this moment as a single period of time, where basically, you can make the pro, the the particular attestation. There's a period of time for that proof to be done, and you're finished. There's no other looking at the build ahead that would come from the optimistic occurring of of new blocks being added.

207

00:35:38.080 --> 00:35:48.029

Jamsheed Shorish: Now we've left one little mini elephant in the room. Which is the question of what do we look at? That joint probability of the proof being unavailable, and the data once asked for being unavailable and incomplete.

208

00:35:48.370 --> 00:35:51.959

Jamsheed Shorish: But we're going to make another assumption and say, it's just a number

209

00:35:52.440 --> 00:36:12.940

Jamsheed Shorish: that's a number of  $P$ . It's some probability that that proofs go bad, that something happens and it goes wrong. Of course, that's something which eventually would be derived from different types of scenarios, of behaviors or accidents, transmission errors, communication errors, computational complexity limits that are hit, attackers that are trying to destroy the system or triggering race mode.

210

00:36:13.110 --> 00:36:25.759

Jamsheed Shorish: All of that's being lumped into this particular value.  $P$ . We break it apart in just a little bit. If we have time, I'll show you kind of in the extension where that can be. Again, reiterating and reusing independence, but in a slightly different fashion.

211

00:36:25.850 --> 00:36:27.939

Jamsheed Shorish: But here we just call it the value of  $P$ .

212

00:36:29.080 --> 00:36:36.769

Jamsheed Shorish: So, as mentioned, you can disentangle this unavailability, unavailability of the proof from the sequence or the approver from the unavailability of the data

213

00:36:36.890 --> 00:36:40.390

Jamsheed Shorish: of a validator pool. But this is something which we don't do yet.

214

00:36:41.840 --> 00:36:54.869

Jamsheed Shorish: Once we have this, we've closed the model. So that was it. That was the entire framework for the model for the single period of economic security. We've got the value of the loss for a validator conditional upon actually coming down to it and getting slashed.

215

00:36:54.980 --> 00:37:04.140

Jamsheed Shorish: If  $\mu$  is the slashing fraction, and  $S$  is the stake. Well, then, the value of the loss when it occurs is just  $\mu$  times  $S$ . These are parameters that can be adjusted.

216

00:37:05.130 --> 00:37:19.640

Jamsheed Shorish: The occurrence probability, the probability of actually having something happen is the probability of being selected

as a validator multiply by the probability of a testing to the block data, and finally, the probability of the proof being unavailable or the data being invalid.

217

00:37:19.640 --> 00:37:42.020

Jamsheed Shorish: what's the product of 3 terms. That 1st term  $N$ . Over  $S$ . Is just the probability of being selected as a validator from a pool of  $S$  sequencers. The second term is this approximation that we did for that  $P$ . Of  $\Omega$ ,  $a$ . That probability of an attestation, and finally, the probability then, of of the proof actually not being available and the data not being possible to show.

218

00:37:44.000 --> 00:38:04.600

Jamsheed Shorish: finally, that was for one validator. That's 1 validator taking the universe is given. So then we multiply by the total population of the of the entire number of sequencers that are there, so that should say, expected loss for population of sequencers.  $S$ . Times the expected loss of each validator. And that's because each sequencer looks forward to the possibility of being elected as a validator.

219

00:38:05.370 --> 00:38:10.670

Jamsheed Shorish: and as a result you end up with the economic security stake as being the product

220

00:38:10.810 --> 00:38:13.490

Jamsheed Shorish: of the probability times. The loss.

221

00:38:15.170 --> 00:38:37.860

Jamsheed Shorish: That's it. That's a number that's a measurement. And the nice thing about this measurement is that it's a thing that can be compared. Once you've specified these types of parameters, you can actually look at the whole scope of the parameters that you can create all of the different types of consensus mechanisms on the basis of varying these parameters. And you'll get something that actually makes sense as a commensurate idea of what's at risk, and therefore what the security is.

222

00:38:39.660 --> 00:38:46.640

Jamsheed Shorish: Okay. I'll stop here quickly and just see if there are any questions or or comments. I'll see if I've missed anything in the chat. Oh, I missed a lot of the chat. I apologize.

223

00:38:46.640 --> 00:38:59.765

Zargham: We're we're talking about kind of tooling and simulations and comment, because this ties into some of what I've seen with the computational folks here at Austrem workshop is the difference between like a

224

00:39:00.210 --> 00:39:08.570

Zargham: like a parameter selection under uncertainty, or like a large scale, robust optimization problem, which is, on one hand more like

225

00:39:09.061 --> 00:39:32.598

Zargham: I would say, like an expert type method versus like the more public engagement, soliciting preferences, developing understanding among stakeholders, which is where I would see more like the streamlit models. And you know, there's obviously a continuum here, and this is a little bit of a tangent from I mean, not really cause. We're talking about engaging with people to understand their preferences over economic mechanisms. But

226

00:39:32.940 --> 00:39:46.029

Zargham: yeah, I think it's it's actually a worthy subject. I don't know whether we wanna further go down that rabbit hole or we wanna even reserve it as a topic for a a future discussion. I'm Rashawn Jacob, there's a

227

00:39:46.140 --> 00:40:00.597

Zargham: I I at least think that a simple thing we could do is make a list of tools and be like this is good for this, you know. Not necessarily great for that. And I even in explaining CAD CAD, I you know, or I was having a conversation with other people about

228

00:40:01.233 --> 00:40:09.586

Zargham: the role of CAD as a tool in a work flow of co-design. Whether it's market or in more institutional problem.

229

00:40:10.337 --> 00:40:34.749

Zargham: That actual co-design process like it merits both demonstrative exploratory models to help people build intuition. It. It merits different models for helping people, maybe do trial and error and get a sense of what they think is good and what might happen if they did it versus again, like, you know, CAD having been designed largely for robust optimization style processes and

230

00:40:34.750 --> 00:41:04.470

Zargham: just like getting people out of the mindset of like, Oh, you know, we use this tool and onto tools. They're useful for certain practices. And you go like, okay, which thing are we doing? Okay? Which tool should we be using? And I, I think the streamlit models are super valuable. And I think there are. Yeah to Shawn's Point Digital Army Armory. Like, we, we actually use a ton of stuff. And we actually have a lot of tacit knowledge about what to use for what and that making that explicit even in just like a table.

231

00:41:04.470 --> 00:41:09.490

Zargham: or like in a you know, small wiki with 6 entries could actually go a really long way.

232

00:41:09.680 --> 00:41:11.839

Zargham: Okay, sorry. And rant.

233

00:41:12.720 --> 00:41:38.400

Jamsheed Shorish: Nice. Yeah, I think that's good, that, yeah. I think it's actually even something that we could leverage and extended the definition of what a concept note is to incorporate those not necessarily for funding opportunities, but really as a way to for the Kms. To surface it and chew on it so that we could ask the Kms. What's the best tool for a 2 week horizon project in which we need to surface a metric, and it could come back and say, Look at Carolina spreadsheet, for example, as our example, that would be a way in which we could say, Okay, let's leverage this information effectively.

234

00:41:38.856 --> 00:41:43.279

Jamsheed Shorish: So just as a plug for for concept notes as a as a pattern.

235

00:41:44.380 --> 00:41:50.100

Zargham: Shawn, did you want to add anything? I realize I I jumped in in response to your messages, and Jacob, as well.

236

00:41:51.190 --> 00:42:03.067

seanmcowen: Can I actually maybe think on it and just put it in the slack channel after? Because I there's a lot of ideas in my head right now, but rather not do half baked like. What if we did this? This this

237

00:42:03.620 --> 00:42:04.150

Zargham: Make sense. I.

238

00:42:04.150 --> 00:42:04.970

seanmcowen: Disappointed.

239

00:42:05.486 --> 00:42:11.410

seanmcowen: No, no, I appreciate it. I actually was going to say something that I was like. I actually would rather

240

00:42:11.470 --> 00:42:17.040

seanmcowen: form this idea a little bit first, st and then propose like, here's some idea of how we might

241

00:42:17.330 --> 00:42:18.360

seanmcowen: do that.

242

00:42:18.360 --> 00:42:35.645

Zargham: Let's make it a thread in slack, so that it's not lost at the end, like. So if you have thoughts, instead of putting it in the even the Zoom chat, putting it in starting a starting, a thread in the it can be in the Econ Working Group Channel. But I think this is worth preserving and coming back to, because I think it's a essentially

243

00:42:36.190 --> 00:42:51.999

Zargham: high benefits to effort ratio things. So we just talk about it. Think about what we want to do, and even make a simple spreadsheet with, you know, some text or something that helps guide guide this, I think we can benefit a lot from doing that. So.

244

00:42:52.240 --> 00:42:54.859

seanmcowen: Yep, yep, I'll I'll start that right after.

245

00:42:55.200 --> 00:42:56.300

seanmcowen: Perfect. Thank you.

246

00:42:58.500 --> 00:42:59.110

Jamsheed Shorish: Print.

247

00:42:59.840 --> 00:43:02.160

Jamsheed Shorish: Okay? Well, in the okay.

248

00:43:02.390 --> 00:43:03.947

Hash: Yeah, I have a question

249

00:43:04.950 --> 00:43:15.860

Hash: the equation that you gave like, I'm looking at it talking. So probability for a validator is the probability of being selected like, would it change if there was a different consensus mechanism, the probability would the probabilities change.

250

00:43:16.610 --> 00:43:40.769

Jamsheed Shorish: It depends upon selection process. Yeah. So basically, Aztec, I said that what they're going to do is they're going to allow anybody that's in the pool of of sequencers and the sequencer set have an equal chance of being selected as a validator every time a block is proposed. So under that technological restriction, then that's essentially the probability is defined. If you were to change how people are selected, then that would naturally change that probability.

251

00:43:41.550 --> 00:43:44.850

Hash: And what's the like? What's the consensus that mechanism that you use?

252

00:43:45.500 --> 00:43:46.799

Hash: Have they shared that.

253

00:43:47.160 --> 00:44:10.146

Jamsheed Shorish: It's pretty much just the idea that they are looking for something where everyone has an equal chance to participate. So if everyone has an equal chance. There's only one distribution. So essentially, you're you've got a  $1$  over  $S$ . Chance of being of being filled in that role. If you weren't filled for the 1st one. You have a  $1$  over  $s$ . Minus one chance for that one, but you have to make sure that they're only being chosen from  $S$  minus one individuals, and so, as you add them all up, it just becomes  $n$  over  $s$ , and so that's

254

00:44:10.410 --> 00:44:15.010

Jamsheed Shorish: the tightest possible number for the most dispersive selection process.

255

00:44:17.090 --> 00:44:28.055

Hash: And think of, they said. For example, they are choosing it

through some kind of a round robin. Okay, one comes after, but then it opens up a hundred's nest of it being given, because now everybody knows what the selection process is.

256

00:44:28.310 --> 00:44:39.579

Jamsheed Shorish: Here. There's nothing like that, because there's nothing to condition on. There's no round Robin. So it's not that everybody gets an equal chance before starting over again. It's possible, although very unlikely, that the same individuals are always being chosen.

257

00:44:41.570 --> 00:44:42.950

Hash: That won't make it to crash.

258

00:44:46.100 --> 00:44:46.760

Jamsheed Shorish: Okay?

259

00:44:47.190 --> 00:45:04.130

Jamsheed Shorish: Well, I promised a comparison to be able to compare to that commitment bond approach. The commitment bond approach is very simple in terms of this definition, under the definition of the economic security that we've come up with. When you only have one entity that has to submit a proof, and they're submitting a certain amount. They're putting this commitment bond up.

260

00:45:04.130 --> 00:45:20.130

Jamsheed Shorish: the probability of a proof being unavailable. Where that's the only thing you're conditioning on again, we're gonna label that as  $P$ , because we don't really care necessarily about the unpredictability or invalidity of a proof data being called on that event,  $w_{\substack{\text{sub} \\ \text{omega} \\ \text{sub} \\ \text{i}}}$ . Is not directly conditioned on.

261

00:45:20.180 --> 00:45:28.640

Jamsheed Shorish: And so the economic security state in that case is just the probability of this particular proof being unavailable, multiplied by what you get slash by.

262

00:45:29.100 --> 00:45:53.659

Jamsheed Shorish: And so we have a simple measurement. That measurement itself still has the ability to be compared. You can still change that parameter  $P$ . In with that particular caveat about  $\Omega$ ,  $I$  not being involved, and it also allows you to compare the magnitudes of the stake that's necessary to compete or to compare 2 values of

this economic security stake that are similar to each other, but may have very different staking behavior in the background.

263

00:45:54.140 --> 00:46:00.750

Jamsheed Shorish: And so, as a stylized example, if you had, these are numbers that are being recommended as initial values currently by Aztec.

264

00:46:01.070 --> 00:46:17.430

Jamsheed Shorish: If you assume that a sequencer has a stake a hundred 1,000 Usd. The commitment bond would be a million dollars Usd. For that single prover that one individual with a commitment bond. Probably the staking slash percentage is 20%. So if you're caught, 20% of that is gone.

265

00:46:17.440 --> 00:46:28.169

Jamsheed Shorish: If the probability of having a essentially proofs go wrong, and not having available data is one out of a hundred chance, and everybody is flipping a coin when they're deciding whether or not to a test.

266

00:46:28.640 --> 00:46:30.500

Jamsheed Shorish: Then, if you've got

267

00:46:30.650 --> 00:46:32.410

Jamsheed Shorish: 25 sequencers.

268

00:46:32.590 --> 00:46:40.860

Jamsheed Shorish: you can have the same roughly, the same economic security as that single prover with a 1 million stake

269

00:46:41.170 --> 00:46:45.880

Jamsheed Shorish: commitment bond commitment. Each of these in this case is around 2,000 Usd.

270

00:46:46.490 --> 00:46:47.290

Jamsheed Shorish: so

271

00:46:47.670 --> 00:47:10.130

Jamsheed Shorish: with 25 validators, what you're putting up as a stake that's available as being at risk is 2.5 million. So of course,

it's in total larger. But you can see that the measurement is based upon the expected loss. We're not just comparing big numbers of being staking versus small numbers. We're actually looking for these mappings of these 2 being commensurate. And of course, as you change S and S. Prime, Mu, PA. And N,

272

00:47:10.400 --> 00:47:31.249

Jamsheed Shorish: these values for the economic security stake are going to change in interesting ways. You're gonna have many different types of methods by which the comparisons are going to be commensurate or not commensurate. So you get an understanding, then, of the way in which this particular economic security can be done. Either more or the same, or less than having a single prover.

273

00:47:34.900 --> 00:47:35.720

Jamsheed Shorish: Okay.

274

00:47:36.540 --> 00:47:38.309

Jamsheed Shorish: that was for a single

275

00:47:39.190 --> 00:47:57.940

Jamsheed Shorish: job like one block that was done. That was the same as when we had the prover commitment bond being just for that one particular block that's showing up, and some of the time it has that we have left we can look at some of the extensions that you can do when we're thinking about this optimistic build ahead. In other words, when you can participate in the creation of another block and another one.

276

00:47:58.260 --> 00:48:00.699

Jamsheed Shorish: and forward up to some time horizon.

277

00:48:02.000 --> 00:48:22.629

Jamsheed Shorish: So let's define that horizon in which we're looking for this time, in which a particular block is acceptable. By T. We have to have a proof that's submitted within that time. T. After which the block data has been attested. In reality, this T is actually called what is what is termed as an epoch, and within which all the blocks themselves are then finalized at once as a particular batch.

278

00:48:22.630 --> 00:48:34.540

Jamsheed Shorish: That means you have different time horizons under

which the proof has to be submitted up to the point at the very last second to last, of those time slots where you have an immediate proof that needs to be submitted. We're abstracting away from that completely.

279

00:48:34.540 --> 00:48:46.199

Jamsheed Shorish: just looking at the expected loss with regard to participation, and not with regard to the question mark of how the probabilities change as that epoch horizon converges to the end of the epoch.

280

00:48:47.470 --> 00:48:53.399

Jamsheed Shorish: So a validator can participate, then, in more than one potentially, but to participate in more than one block proposal.

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00:48:53.680 --> 00:49:11.470

Jamsheed Shorish: And Aztecs decision is kind of like, well, what will happen to a validator after they're slashed. Are they really going to be allowed to participate? There's a potential possibility of doing so. We were just discussing in the internal. Jacob gave us the information that it's unlikely. But we can still consider 2 cases, one of which is a permissive environment where

282

00:49:11.470 --> 00:49:25.500

Jamsheed Shorish: the validator is completely allowed to continue to participate in future proposals over that time horizon. In other words, you've been slashed bad. You were very bad, but come on back in. You have a chance to be a validator next time if your number comes up and you have a chance to participate

283

00:49:26.000 --> 00:49:32.620

Jamsheed Shorish: the more realistic one which appears to be the one that Aztec is moving toward, or is expected to be moving toward is a more restrictive

284

00:49:32.870 --> 00:49:43.840

Jamsheed Shorish: set of consequences. Where you jail or you ban the validator once they've done something wrong. Then you say, Nope, you're not allowed to participate in the the future proposals for at least that period of time.

285

00:49:45.040 --> 00:49:49.020

Jamsheed Shorish: and you can compare them this permissiveness versus

restrictive validator treatment.

286

00:49:49.060 --> 00:50:01.580

Jamsheed Shorish: Intuitively, you could think about the idea of jailing and banning as somehow, because you're extracting bad behavior from the system. You expect the quality of service to go up, but for individual validators sitting on the sidelines means not being at risk.

287

00:50:02.520 --> 00:50:15.439

Jamsheed Shorish: And so there is this trade-off that's occurring. This is where it's a nice metric to separate out things like incentivizing individuals, to do the right thing from the actual assessment of is the system economically secure.

288

00:50:15.460 --> 00:50:43.659

Jamsheed Shorish: So here we're making a shark distinction. We're not confounding the 2 and saying, Oh, it's also incentivization of this particular behavior. Well, no, we have no idea in this particular environment, for example, whether or not individuals who are validators receive a reward, and we don't want to model that because we don't know what that's going to look like. So we've put that completely to one side, and that means that this is augmented by and can in principle change with incentive mechanisms that are designed to ensure that they like what they're doing as validators.

289

00:50:46.160 --> 00:50:56.909

Jamsheed Shorish: Okay, so we have what might be called horizon dependent economic security. Now that we introduce this notion of being able to participate potentially in future blocks that are being created.

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00:50:57.370 --> 00:51:20.980

Jamsheed Shorish: that means that we can kind of extend a little bit, the idea of this proof being unavailable versus the idea of this proof being invalid to allowing an independence, assumption of M. And Q. This in itself could have been done earlier. I've just put it in here because it allows us to kind of build a little bit more rich tapestry in the in the event that we can actually converge everything to under a single period, we can still leverage this this universe.

291

00:51:21.607 --> 00:51:25.909

Jamsheed Shorish: If we have the simplification, then the probability of loss we can define as

292

00:51:25.920 --> 00:51:34.469

Jamsheed Shorish: the product of those probabilities that we had before. That's essentially the same thing we saw before, just written now with the the definition or denotation of you

293

00:51:35.260 --> 00:51:43.040

Jamsheed Shorish: for the permissive treatment of the environment, the validator is always available for these future block proposals, and so their economic security is essentially

294

00:51:43.090 --> 00:51:47.929

Jamsheed Shorish: adding up the amount that that's risk at each period in time that they could be

295

00:51:48.070 --> 00:51:49.270

Jamsheed Shorish: participating

296

00:51:50.370 --> 00:51:59.139

Jamsheed Shorish: by contrast for the restrictive treatment. The validator itself may not be available from the succeeding block proposal. So you have to take a look at. Okay? Were they slashed?

297

00:51:59.280 --> 00:52:12.860

Jamsheed Shorish: Yes, then they cannot participate, but if they weren't slashed, then they could try again. Are they slashed at that point? If yes, they're excluded, if not, then they can continue again. And so you end up with this little infinite, not infinite, a finite sequence that you can then.

298

00:52:12.910 --> 00:52:22.009

Jamsheed Shorish: right in a nice converge state to say, Okay, in this particular setting. The economic security state, then, has the ability to be expressed as a part of that probability.

299

00:52:22.010 --> 00:52:46.169

Jamsheed Shorish: And here is where the number of sequences actually matters, you actually do have an individual value. Where now that  $S$  over  $N$  part where we actually had the cancellation from  $N$  over  $S$ , where the  $S$ 's were cancelling. That doesn't occur anymore. And you actually have scaling with regard to the number of sequences. So the numbers, the sequences, the larger the sequencer pool. All things

equal you're gonna find an interior point where you have like an optimal number of sequencers.

300

00:52:46.940 --> 00:53:14.430

Jamsheed Shorish: Now, if you've got in a system, where not everybody is. Where you're not asking for a size of the Validator Pool to be more than half of the number of sequencers. Then it's the case that your expected loss that would occur when you are able to continually come back. Every single period is higher than the expected loss when you're actually being forced to sit on the sidelines. That just is this idea that the the restrictive treatment lowers the economic security stake.

301

00:53:15.160 --> 00:53:25.520

Jamsheed Shorish: It's clearly the case that you're looking at a higher expected loss when you actually are repeating things over many periods than just one. So that's just a trivial result. But it's consistency check.

302

00:53:25.740 --> 00:53:47.839

Jamsheed Shorish: And as a restriction on  $T$  you would like to be able to ensure. And this is really for just making sure that the numbers work out as integers. That, you want to choose a horizon which is at least as large as this multiple that is described as  $S$  over  $N$ , if you have something which is just,  $T$  equals one, and the single period analysis itself is, is up being applied.

303

00:53:49.150 --> 00:53:50.030

Jamsheed Shorish: Okay.

304

00:53:50.580 --> 00:54:10.369

Jamsheed Shorish: any questions about the horizon. This is a simplification, because it doesn't look at that exploding like exploding offer at at the end of this epoch, then the blocks which are being proposed right before the end have basically no time for approve. And so part of the the extension of this to a real model, a full model would be the time taken, the clock that's ticking

305

00:54:10.390 --> 00:54:20.280

Jamsheed Shorish: for proof to be submitted, because you know that if you're getting very close to the end of that time period. It's possibly unlikely that a proof is going to show up, that's all being abstracted away from.

306

00:54:24.830 --> 00:54:25.700

Jamsheed Shorish: Okay

307

00:54:26.336 --> 00:54:40.683

Jamsheed Shorish: the last extension that I'll just treat. This is one other piece of realism. So these are all pieces that eventually will find their way into that spreadsheet. That's a lot of next week's work. On the books is idiosyncratic voting. This is the idea.

308

00:54:40.970 --> 00:54:41.640

jhack: Like, click.

309

00:54:41.870 --> 00:54:44.629

jhack: Yeah. Sorry. Sorry. I. I was still thinking about the

310

00:54:44.850 --> 00:54:56.239

jhack: intuition that restrictive treatment of validators lowers the economic security stake because the economic security stake only applies to the let's say, 300 validators that are selected from the sequencer pool

311

00:54:56.816 --> 00:55:10.540

jhack: and then a new 300 can be selected for the next epoch after somebody gets jailed. Right? So it should always be at these 300. I'm I'm not exactly sure why the low or is this the economic Security State.

312

00:55:10.810 --> 00:55:18.140

Jamsheed Shorish: Right. So the idea here is to not treat this as validators that are being conditioned upon having already become a validator.

313

00:55:18.180 --> 00:55:29.000

Jamsheed Shorish: If you do that, then you are looking at only the case of whether or not I was elected. And then what the expectation is, what's actually happening for the economic security state is that you're a sequencer.

314

00:55:29.340 --> 00:55:33.260

Jamsheed Shorish: And you're saying, what's the probability of me

actually being selected

315

00:55:33.360 --> 00:55:35.400

Jamsheed Shorish: as a validator first? st

316

00:55:36.480 --> 00:55:46.529

Jamsheed Shorish: And so if you have an epoch where you're banned for one period, then what happens in that particular horizon? Remember, this is this is a an economic security that's bounded by that horizon.

317

00:55:46.590 --> 00:55:48.290

Jamsheed Shorish: Within that horizon

318

00:55:48.370 --> 00:55:50.609

Jamsheed Shorish: the economic security stake falls

319

00:55:50.890 --> 00:55:53.310

Jamsheed Shorish: if you're sitting on the sidelines for most of it.

320

00:55:53.390 --> 00:56:01.200

Jamsheed Shorish: because your individual amount that you would be receiving is low, and that's true for every single sequencer that's waiting on the wings, whether or not to jump in.

321

00:56:01.520 --> 00:56:03.300

Jamsheed Shorish: It doesn't say, however.

322

00:56:03.570 --> 00:56:14.169

Jamsheed Shorish: that when you come back the following epoch, and you start over again. And all this forgiven, that you necessarily don't necessarily have an accumulation path. It just depends upon how much of that accumulation would be

323

00:56:14.210 --> 00:56:18.480

Jamsheed Shorish: compared to the permissive environment where you are continually able

324

00:56:18.510 --> 00:56:20.849

Jamsheed Shorish: to be exposed to getting slashed.

325

00:56:21.866 --> 00:56:22.340

jhack: That's fine.

326

00:56:22.340 --> 00:56:23.360

Jamsheed Shorish: Reference. Book, here.

327

00:56:23.530 --> 00:56:24.360

jhack: Got it. Got it?

328

00:56:24.650 --> 00:56:25.310

jhack: Yep.

329

00:56:25.560 --> 00:56:36.470

Jamsheed Shorish: And that's and that's a good point. That's a good point to to what I glossed over in some of the approximation. Is this this idea that it's the sequencer that's sitting there saying, what's gonna happen to me.

330

00:56:36.910 --> 00:56:49.199

Jamsheed Shorish: you know, in a sense, a sequencer in the sequencing pool may not want to be a validator, but they're on the hook for being a validator that maybe they can opt out. Maybe there is some restricted set that we'd be looking at in reality. But, exanti.

331

00:56:49.220 --> 00:56:54.289

Jamsheed Shorish: Every sequencer has the chance of being drafted into the validator pool and then being slashed.

332

00:56:55.180 --> 00:57:00.440

Jamsheed Shorish: And that's the the way in which this economic security stake makes that precise in that fashion.

333

00:57:04.160 --> 00:57:04.800

Jamsheed Shorish: Other questions so.

334

00:57:04.800 --> 00:57:09.810

Hash: So is it for all. For every different epoch new set of validators are chosen that they have with.

335

00:57:09.810 --> 00:57:11.340

Jamsheed Shorish: Every sequence is locked.

336

00:57:11.400 --> 00:57:18.030

Jamsheed Shorish: every block. So there are, let's say, 50 time slots for blocks to be proposed within an epoch.

337

00:57:18.250 --> 00:57:21.729

Jamsheed Shorish: Then at every single one of those time slots of those 50

338

00:57:21.860 --> 00:57:26.459

Jamsheed Shorish: there is a new validator set that's drawn from the population of sequencers.

339

00:57:28.720 --> 00:57:37.369

jhack: That is the ideal scenario, by the way, but not what Aztec has yet decided on. But it's something that we would recommend in any case, because otherwise you get into weird

340

00:57:37.520 --> 00:57:39.100

jhack: strategic behaviors as well.

341

00:57:39.330 --> 00:57:39.990

jhack: Yep.

342

00:57:39.990 --> 00:57:49.260

Jamsheed Shorish: Yeah, it's a it's an it's an ex anti suggestion that that initially came from from Aztec about how are they thinking about the the selection process occurring. Yeah, that's right.

343

00:57:51.145 --> 00:57:58.645

Carolina: Just, you know, to understanding practice what would happen. So now we're not considering, you know, the browser distribution.

344

00:57:59.330 --> 00:58:04.719

Carolina: the failure, probably distribution given. You know the the time that is left until we reach P.

345

00:58:05.190 --> 00:58:09.399

Carolina: But let's say, let's say that you know.

346

00:58:09.980 --> 00:58:11.770

Carolina: 49 out of

347

00:58:11.960 --> 00:58:29.579

Carolina: 50 blocks were able to be successful. Right? There's only one that it's that it's just that fail, for some reason. So the entire batch would be slashed. You know, kind of like that. The team would go back to 50, minus to to whatever it was minus 50.

348

00:58:29.910 --> 00:58:33.140

Carolina: Like all the work was wasted basically.

349

00:58:33.740 --> 00:58:39.780

Jamsheed Shorish: Yeah. But reorganization occurs whenever it gets an invalid. Anything that was built on something coming later.

350

00:58:40.200 --> 00:58:44.479

Jamsheed Shorish: By the time that proof isn't available all that has to be unwound. That's correct.

351

00:58:45.410 --> 00:58:49.430

Carolina: So the work for the 49 that we're correct would need to be repeated again

352

00:58:49.680 --> 00:58:52.480

Carolina: until the 50 at once are valid.

353

00:58:53.400 --> 00:58:55.339

Carolina: Oh, wow! Okay, I see.

354

00:58:55.730 --> 00:58:58.109

Jamsheed Shorish: So it's very costly in that. In that respect, especially for things.

355

00:58:58.110 --> 00:58:58.510

Carolina: The i.

356

00:58:58.510 --> 00:59:00.160

Jamsheed Shorish: Magnus. Yeah, that's right.

357

00:59:01.310 --> 00:59:02.179

Jamsheed Shorish: and are the but.

358

00:59:02.180 --> 00:59:02.960

Hash: Proofs

359

00:59:03.230 --> 00:59:08.869

Hash: generated for every block or once, every epoch. What was the cadence? There.

360

00:59:09.290 --> 00:59:11.530

Jamsheed Shorish: It's in in its operational

361

00:59:11.660 --> 00:59:16.919

Jamsheed Shorish: setting. And of course Jacob can correct me if I'm wrong, but it's essentially batched at the end of the epoch.

362

00:59:16.960 --> 00:59:30.430

Jamsheed Shorish: So there will be one proof that is actually rolled up all of the individual proofs for all of the blocks, so that you actually send one of them at the end of the epoch, the idea being that you have a more efficient method of then transmitting this to l 1 for finalization.

363

00:59:32.290 --> 00:59:45.130

jhack: That's correct. That's essentially like a merkel tree where you roll up to one hash and originally did did this per block. But now to speed up. They're thinking of batching blocks together to have one proof per batch of blocks.

364

00:59:48.950 --> 00:59:50.740

Hash: Wow! That's I mean, if.

365

00:59:50.800 --> 01:00:01.479

Hash: say, it's a 1 h window, like every batch, is 1 h. So one proof goes wrong or one transaction. There's some hanky panky somewhere and everything, so they'll have to go all the way back

366

01:00:01.860 --> 01:00:04.660

Hash: to maybe thousands of transactions back. That is like.

367

01:00:04.660 --> 01:00:11.399

jhack: Luckily they're thinking of like 6 to 12 min, or something like that. So something like a bitcoin block that gets rolled up again?

368

01:00:11.860 --> 01:00:13.340

jhack: Or did it get to New York.

369

01:00:14.850 --> 01:00:15.570

Hash: Okay.

370

01:00:18.740 --> 01:00:36.449

Jamsheed Shorish: Alright in the time left. I don't wanna go into too much. I wanna leave a little more time for some sort of you know general conversation or assessment of this, but I do wanna show you very quickly that with idiosyncratic voting the system simply passes through just the the derivations of the distributions of the individual votes changes

371

01:00:36.450 --> 01:01:01.770

Jamsheed Shorish: when everybody has their own little probability of making a vote or not making a vote of an attestation you would do end up with kind of a heinous mess X. Ante. On what that particular set of combinations is, because it's much richer now that everybody has their own chance. But fortunately there's a bound that exists even in this case, and it's fairly simple to derive. It's usefulness, of course, depends a lot upon the parameters that you have.

372

01:01:03.050 --> 01:01:15.490

Jamsheed Shorish: but nothing changes. The definition of the economic security state doesn't change the way in which you compute the probabilities conditional upon our assumptions doesn't change. It's just the richness of the underlying probability, that is, that is being set up.

373

01:01:15.490 --> 01:01:27.890

Jamsheed Shorish: and a further relaxation. And this is something that I'm I'm looking at is how much time will I have next week to to kind of investigate versus making sure that the that we're good to go for the tool and the and the the documentation for this little exercise

374

01:01:28.160 --> 01:01:50.619

Jamsheed Shorish: is to say, Well, hey, if I'm an individual, and I'm sitting there as a sequencer, and I'm looking at everybody else around me. I may not actually know any of these. Ais but I may have some idea about the distribution is, maybe there's a uniform distribution of probabilities, or there's a concentration to subpopulations. Exist. There are a bunch of people that I happen to know are very good. From prior experience. My priors about their behaviors are very high.

375

01:01:50.620 --> 01:02:12.000

Jamsheed Shorish: so I could set up a Bayesian probability distribution that is reinforced on the successes or failures of the voting process. So you can chain together some very nice, albeit more complex, dependencies upon what you're seeing as data. And then actually, as you're participating as a validator, you're almost a sensor receiving the result of the vote and Updating your idea about the rest of the world accordingly.

376

01:02:12.000 --> 01:02:26.670

Jamsheed Shorish: That can also be placed into this context, and would be of interest to see if particular failure modes of voting where you're voting for the wrong thing, or you're voting for something which clearly is not going to succeed, or you're seeing a cartel of individuals doing something nefarious.

377

01:02:26.670 --> 01:02:31.719

Jamsheed Shorish: These would then have some some implications. But that's not something that can be done within the time. Horizon

378

01:02:33.020 --> 01:02:33.770

Jamsheed Shorish: firm.

379

01:02:35.100 --> 01:02:39.179

Jamsheed Shorish: Well, thanks. Any final questions or thoughts.

380

01:02:41.920 --> 01:02:52.090

Carolina: Yeah. So I actually just wanted to. In terms of the spreadsheet, for example, to to show the the change in time for the aggregated loss.

381

01:02:52.700 --> 01:02:57.979

Carolina: So can you? Sorry? Can you go back to the slide? Maybe. Oh, you've already stop. Sorry

382

01:02:58.610 --> 01:03:00.559

Carolina: and let you have the.

383

01:03:05.740 --> 01:03:14.015

Carolina: I think it is one minus you the power of. Can you go back? Oh, sorry. I have lots of stops.

384

01:03:15.040 --> 01:03:15.790

Jamsheed Shorish: Problem here.

385

01:03:15.790 --> 01:03:28.039

Carolina: That one down. Exactly. So. I I see that this is defined by you know, for some  $T$  capital  $T$ . But if you were to split this into, you know every

386

01:03:28.350 --> 01:03:30.369

Carolina: every time. Step

387

01:03:31.020 --> 01:03:32.150

Carolina: I'll eat

388

01:03:33.280 --> 01:03:40.420

Carolina: then, what? What would be the the logic here for that if you were to, or this will be just of of when

389

01:03:40.600 --> 01:03:47.542

Carolina: number, and that's it. Because I I believe that right now, what we're trying to do is to kind of like, see the progression of this

390

01:03:47.910 --> 01:03:50.529

Carolina: economic security stake over time.

391

01:03:50.670 --> 01:03:51.790

Carolina: And what?

392

01:03:52.040 --> 01:03:56.039

Carolina: But here you have no capital fee. And that's it, right. Seems that it's just one number.

393

01:03:56.480 --> 01:04:05.349

Jamsheed Shorish: So what's happening here exactly is you're you're essentially reducing the entire chain of what might occur over 50 periods into one expectation at the beginning.

394

01:04:05.850 --> 01:04:19.380

Jamsheed Shorish: So essentially, it reduces those that those 50 rows down to essentially a single row that says, projecting forward, what will happen to me because I can exploit the probability of being slashed and then being thrown out.

395

01:04:19.750 --> 01:04:47.320

Jamsheed Shorish: And so that gives me a closed form solution. So instead of having a numerical process where I'm following this across. Since I know that when I get slashed I have to sit out at the very beginning of those 50 periods. Let's say I can compute the single number that occurs for that, and of course you can vary T. So you could have 49 more rows where T is going down by one each time, and that's giving you the chance of well, if I only have 48 periods left! If I only have 47 periods left, if I only have 2 periods left.

396

01:04:47.440 --> 01:04:54.499

Jamsheed Shorish: and that may give us a little bit of a proxy for that notion of what happens when that big horizon is coming closer and closer to us.

397

01:04:54.510 --> 01:05:05.890

Jamsheed Shorish: So that will be, yeah, I think, in the end. And that's something we can discuss, you know, in the spreadsheet we could still have those 50 rows. But we would basically be counting down from T equals. Cap T. Equals 50 down to Cap T equals one.

398

01:05:07.450 --> 01:05:12.625

Carolina: Okay, I see what you mean. But then, can you remind me with you with the

399

01:05:13.070 --> 01:05:14.210

Carolina: this steak?

400

01:05:16.250 --> 01:05:20.419

Carolina: and S. No,  $\mu$  is the this, this state S is the percent.

401

01:05:20.420 --> 01:05:25.540

Jamsheed Shorish: Slashing the slashing fraction. So mute times, S. Is how much you lose when you're when you're slashing.

402

01:05:25.540 --> 01:05:26.960

Carolina: Yes, and you

403

01:05:27.350 --> 01:05:28.509

Carolina: that is mute, and you.

404

01:05:28.510 --> 01:05:31.990

Jamsheed Shorish: Probability of being caught in a slashing event.

405

01:05:32.490 --> 01:05:32.910

Carolina: Okay.

406

01:05:32.910 --> 01:05:33.370

Jamsheed Shorish: That's here.

407

01:05:33.370 --> 01:05:36.080

Carolina: I see. Thank you. Perfect. Yeah.

408

01:05:36.990 --> 01:05:37.580

Jamsheed Shorish: Hmm.

409

01:05:37.930 --> 01:05:42.449

Jamsheed Shorish: it's just rewrites just using a different piece of notation. I didn't want to keep yeah carrying around the same thing. Yeah.

410

01:05:42.450 --> 01:05:43.865

Carolina: Okay. Cool no worries.

411

01:05:44.610 --> 01:06:07.080

Jamsheed Shorish: But the idea here is, and that's kind of links back

to what? What Jacob was saying is this this idea that what we're looking for is, you know, something that we don't really know whether or not that that T is an optimal thing that can be chosen, or if it's something that just is hard coded. I think they're looking also for guidance on how big T should be. And so this is maybe one way in which we can. We can examine that question. Within a spreadsheet, of course. And give them some guidance on that.

412

01:06:09.170 --> 01:06:10.180

Carolina: Okay, thank you.

413

01:06:10.540 --> 01:06:11.500

Carolina: Sure, that's good.

414

01:06:13.100 --> 01:06:15.121

Jamsheed Shorish: Alright. Well, I know we're at time

415

01:06:15.670 --> 01:06:40.189

Jamsheed Shorish: any final thoughts or questions, just to kind of hopefully with regard to the the numbers and everything no big deal. But just with regard to the understanding of how do we distill a very quick turnaround? Ask into something which is usable, which is useful and usable and useful, comes from from our side, saying, Hey, this is useful but usable, coming from their side, saying, Okay, we need a tool where we can actually use this quickly. This is part of the process kind of the sausages made.

416

01:06:45.540 --> 01:06:46.290

Jamsheed Shorish: Pardon me.

417

01:06:46.990 --> 01:06:47.770

Jamsheed Shorish: okay.

418

01:06:48.190 --> 01:06:51.550

Hash: For the 1st time I could understand from

419

01:06:51.580 --> 01:06:53.910

Hash: like the 0 min all the way to

420

01:06:53.990 --> 01:07:00.902

Hash: the last minute, like I got everything and this like in 2 years?

Is this been the 1st time? So yeah, I'm happy.

421

01:07:01.500 --> 01:07:06.299

Jamsheed Shorish: Good. I'm glad to hear it. Of course the recordings are are always made and available for for anybody else

422

01:07:06.330 --> 01:07:09.040

Jamsheed Shorish: wants to peruse, and I'll probably go through it again as well.

423

01:07:10.490 --> 01:07:11.350

Jamsheed Shorish: Well, thank you. So.

424

01:07:11.350 --> 01:07:12.000

Carolina: So much.

425

01:07:12.150 --> 01:07:13.050

Zargham: That was great.

426

01:07:13.760 --> 01:07:14.350

Jamsheed Shorish: Grit.

427

01:07:14.650 --> 01:07:15.240

peterhacker: Thanks for shit.

428

01:07:15.240 --> 01:07:15.686

jhack: Thank you.

429

01:07:16.350 --> 01:07:16.969

David Sisson: You take guys.

430

01:07:16.970 --> 01:07:17.859

Carolina: Mind. Thank you.

431

01:07:18.200 --> 01:07:18.860

jhack: But.